

Lab one: Basic Shell Commands and Bash Usage in RHEL 9

Section 1: Logging in and Accessing the System

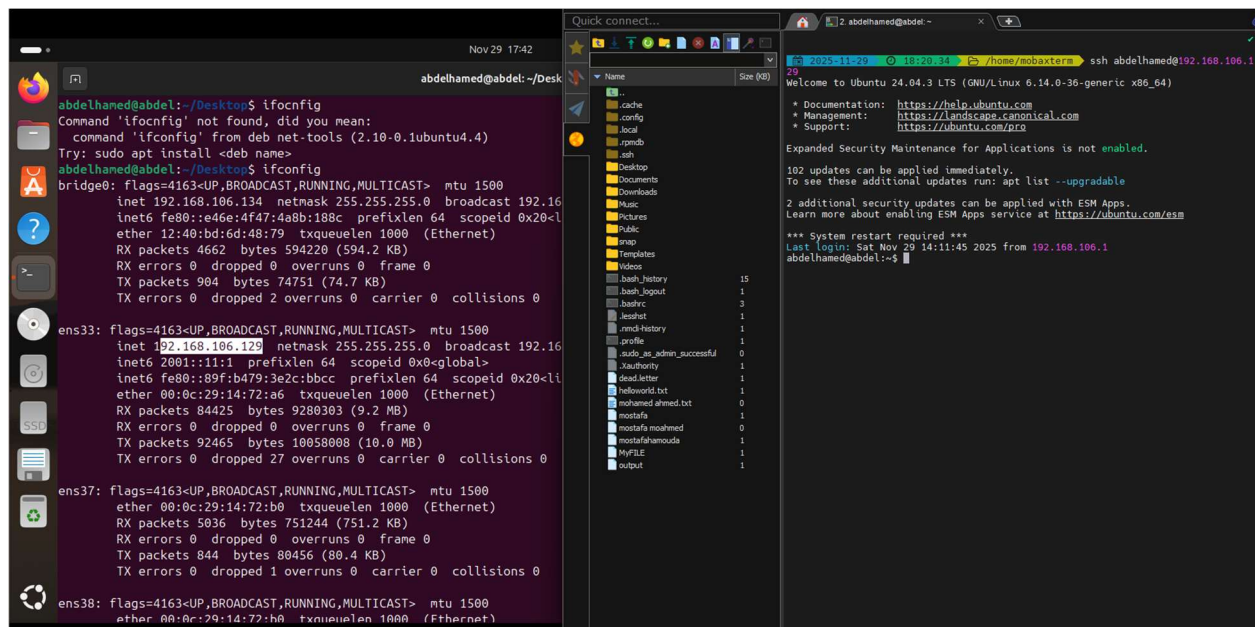
1. Login Task (GUI):

- Open the terminal in the GUI of your RHEL 9 system.
- **Question:** What command do you use to open a terminal in RHEL via the GUI?

Answer: `ctrl + Alt + t`

2. Login Task (SSH):

- From another system, SSH into the RHEL machine.
- **Question:** What command did you use to SSH into the system, and what was the IP address of your RHEL machine?
- **Answer:**



The screenshot displays a desktop environment with a terminal window and an SSH session window. The terminal window shows the execution of the `ifconfig` command, which is not found, followed by the installation of `net-tools` using `sudo apt install <deb name>`. The `ifconfig` command is then executed, showing network interface details for `bridge0`, `ens33`, and `ens38`. The SSH session window shows the user `abdelhamed` logging in to a system running Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-36-generic x86_64). The terminal window also shows the output of the `ifconfig` command for the `ens33` and `ens38` interfaces.

```
abdelhamed@abdel:~/Desktop$ ifconfig
Command 'ifconfig' not found, did you mean:
  command 'ifconfig' from deb net-tools (2.10-0.1ubuntu4.4)
Try: sudo apt install <deb name>
abdelhamed@abdel:~/Desktop$ ifconfig
bridge0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
        inet 192.168.106.134 netmask 255.255.255.0 broadcast 192.168.106.255
        inet6 fe80::e46e:4f47:4a8b:188c prefixlen 64 scopeid 0x20<link>
        ether 12:40:bd:6d:48:79 txqueuelen 1000 (Ethernet)
        RX packets 4662 bytes 594220 (594.2 KB)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 904 bytes 74751 (74.7 KB)
        TX errors 0 dropped 2 overruns 0 carrier 0 collisions 0

ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
        inet 192.168.106.129 netmask 255.255.255.0 broadcast 192.168.106.255
        inet6 2001::11:1 prefixlen 64 scopeid 0x0<global>
        inet6 fe80::89f:b479:3e2c:bbcc prefixlen 64 scopeid 0x20<link>
        ether 00:0c:29:14:72:a6 txqueuelen 1000 (Ethernet)
        RX packets 84425 bytes 9280303 (9.2 MB)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 92465 bytes 10058008 (10.0 MB)
        TX errors 0 dropped 27 overruns 0 carrier 0 collisions 0

ens37: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
        ether 00:0c:29:14:72:b0 txqueuelen 1000 (Ethernet)
        RX packets 5036 bytes 751244 (751.2 KB)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 844 bytes 80456 (80.4 KB)
        TX errors 0 dropped 1 overruns 0 carrier 0 collisions 0

ens38: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
        ether 00:0c:29:14:72:b0 txqueuelen 1000 (Ethernet)
```

```
2025-11-29 18:20:34 /home/mobaxterm ssh abdelhamed@192.168.106.1
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-36-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/support

Expanded Security Maintenance for Applications is not enabled.

102 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***
Last login: Sat Nov 29 14:11:45 2025 from 192.168.106.1
abdelhamed@abdel:~$
```

- To change password of user use command:

```
abdelhamed@abdel:~$ passwd
Changing password for abdelhamed.
Current password:
New password:
Retype new password:
passwd: password updated successfully
abdelhamed@abdel:~$ █
```

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Section 2: Understanding Bash Shell

3. Check Default Shell:

- Run the command to check which shell you are currently using.
- **Question:** What is the command to check the current shell? What output did you get?

```
abdelhamed@abdel:~$ ps
  PID TTY          TIME CMD
 10297 pts/3    00:00:00 bash
 10456 pts/3    00:00:00 ps
abdelhamed@abdel:~$ █
```

4. Bash Commands:

- Run the command `echo $SHELL` to ensure you're using the bash shell.
- **Question:** Is `/bin/bash` shown in your output? If not, which shell are you using?
- **Answer :** yes I'm using the bash shell , PTS is for SSH

Section 3: Running Simple Commands

5. Working Directory:

- Use a command to check your current working directory.
- **Question:** What command did you use, and what directory are you in right now?

```
abdelhamed@abdel:~$ pwd
/home/abdelhamed
abdelhamed@abdel:~$
```

- **Answer :**

6. List Files:

- Run a command to list all files in the /home directory.
- **Question:** Which files and folders are present in /home?
- **Answer :**

```
abdelhamed@abdel:/$ ls
bin                home               NewTemp           snap              usr
bin.usr-is-merged  lib               opt              softlinks        var
boot              lib64             proc             srv              work
cdrom             lib.usr-is-merged root             swap.img
dev              lost+found        run             sys
dir             media            sbin           Temporary
etc            mnt             sbin.usr-is-merged tmp
```

7. Display Date:

- Run a command to show the current date and time.
- **Question:** What is the output of the date command?

```
abdelhamed@abdel:/$ date
Sat Nov 29 06:27:27 PM EET 2025
abdelhamed@abdel:/$
```

- **Answer**

Section 4: Learning Command Syntax

8. Command with Options:

- Use the ls command with options to display detailed information about the files in /etc.
- **Question:** What is the command you used to list files in long format? How many files or directories are listed in /etc?

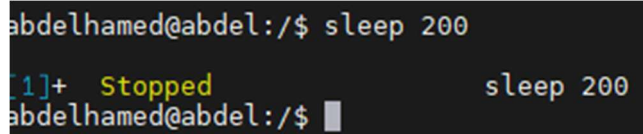
```
abdelhamed@abdel:/$ ls -ll | wc -l
32
abdelhamed@abdel:/$
```

- **Answer :**

Section 6: Using Terminal Shortcuts

9. Interrupt Process:

- Run the sleep 100 command, then use a keyboard shortcut to stop the process.
- **Question:** Which shortcut did you use to interrupt the sleep command?

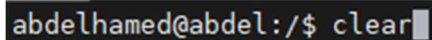
A terminal window with a dark background. The prompt is 'abdelhamed@abdel:/\$'. The command 'sleep 200' has been entered. A new line shows '[1]+ Stopped sleep 200'. The prompt is now 'abdelhamed@abdel:/\$' followed by a cursor.

```
abdelhamed@abdel:/$ sleep 200
[1]+  Stopped                  sleep 200
abdelhamed@abdel:/$
```

- **Answer :**

10. Clear Terminal:

- Use the shortcut to clear the terminal screen.
- **Question:** What is the shortcut to clear the terminal?

A terminal window with a dark background. The prompt is 'abdelhamed@abdel:/\$'. The command 'clear' has been entered, and the cursor is at the end of the line.

```
abdelhamed@abdel:/$ clear
```

- **Answer :**

11. Command Autocompletion:

- Type hos in the terminal and press Tab to auto-complete it to hostname.
- **Question:** Did the Tab key auto-complete the command for you?
- Answer: it will not complete because there is more than one command start with host so the shell will get u the desired output u want to use

```
abdelhamed@abdel:/$ host
host      hostid    hostname  hostnamectl
abdelhamed@abdel:/$ host
```

- **Answer:**

Bonus Tasks

12. Command History:

- Use the keyboard shortcut to browse through the previous commands you've entered.
- **Question:** What shortcut allows you to view and re-run previous commands?
- **Answer :** History, u can use the arrow up and down to navigate through u r history + u can use ctrl + r to change and use one of the used history

13. Log Out:

- Use a terminal shortcut to log out of your current session.
- **Question:** Which shortcut did you use to log out of the session?
- **Answer :** ctrl+D or u can write exit